Rings That Force Their Modules: Ring-Theoretic Conditions Implying That Every Module Has a Prescribed Type

Abstract

A module is the "container" in which a ring acts. We study the rings R for which *every* left R-module is of one fixed type: free, projective, injective, or flat. In each case the module-theoretic condition is equivalent to a purely internal condition on R: being a division ring, semisimple, or von Neumann regular. We give complete proofs of the four classical characterizations and collect, with references, the further implications of the same kind (quasi-Frobenius, hereditary, perfect rings).

1 Conventions

All rings are associative with $1 \neq 0$, and modules are unital left modules unless stated otherwise. A module is *simple* if it is nonzero with no proper nonzero submodules, and *semisimple* if it is a direct sum of simple modules. Equivalently, a module is semisimple if and only if every submodule is a direct summand. We say that a ring R *forces* a property $\mathcal{P}$ of modules if every R-module has $\mathcal{P}$.

Definition 1. A ring R is *semisimple* if ${}_RR$ is a semisimple module. R is *von Neumann regular* if for every $a \in R$ there is $x \in R$ with $a = axa$.

2 Forcing free modules

Theorem 2. *Every left R-module is free if and only if R is a division ring.*

Proof. ($\Leftarrow$) Every module over a division ring is a vector space, and every vector space has a basis (Zorn's lemma).

($\Rightarrow$) Since $R \neq 0$, Zorn's lemma gives a maximal left ideal $\mathfrak{m}$, and $R/\mathfrak{m}$ is a simple module. By hypothesis $R/\mathfrak{m} \cong R^{(B)}$ for some set B, which is nonempty since $R/\mathfrak{m} \neq 0$. A free module on two or more generators has a proper nonzero submodule, so $|B| = 1$ and $R \cong R/\mathfrak{m}$. Thus ${}_RR$ is simple, i.e. the only left ideals of R are 0 and R.

Let $a \neq 0$. Then $Ra = R$, so $ba = 1$ for some b. Also $b \neq 0$, so likewise $cb = 1$ for some c. Then $c = c(ba) = (cb)a = a$, so $ab = 1$. Thus every nonzero element is a unit and R is a division ring. $\square$

3 Forcing projective or injective modules

Theorem 3. *For a ring R the following are equivalent.*

(i) *R is semisimple.*

(ii) *Every left R-module is projective.*

(iii) *Every left R-module is injective.*

(iv) *Every left R-module is semisimple.*

(v) *Every short exact sequence of left R-modules splits.*

Proof. (i)⇒(iv). A free module $R^{(I)}$ is a direct sum of copies of a semisimple module, hence semisimple. Every module is a quotient of a free module, and a quotient of a semisimple module is semisimple (it is the image of a direct sum of simples, and each simple maps either to zero or isomorphically onto its image, so the image is a sum of simples; a sum of simple submodules is a direct sum of a subfamily).

(iv)⇒(v). In a semisimple module every submodule is a direct summand, so for $0 \to A \xrightarrow{f} B \to C \to 0$ the submodule $f(A) \subseteq B$ has a complement and the sequence splits.

(v)⇒(ii),(iii). Let P be any module and choose an epimorphism $F \to P$ with F free, with kernel K. By (v) the sequence $0 \to K \to F \to P \to 0$ splits, so P is a direct summand of F and hence projective. Likewise, any module E embeds in an injective module E', and the sequence $0 \to E \to E' \to E'/E \to 0$ splits, so E is a direct summand of E' and hence injective.

(ii)⇒(i). Let $I \subseteq R$ be a left ideal. The module R/I is projective, so $0 \to I \to R \to R/I \to 0$ splits and I is a direct summand of ${}_RR$. Thus every left ideal is a summand, i.e. ${}_RR$ is semisimple.

(iii)⇒(i). Let $I \subseteq R$ be a left ideal. Then I is injective, so the inclusion $I \hookrightarrow R$ splits, and again I is a direct summand of ${}_RR$. □

Example 4. $\mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$ is semisimple, so all its modules are projective and injective. By contrast $\mathbb{Z}/4\mathbb{Z}$ is not semisimple: the module $\mathbb{Z}/2\mathbb{Z}$ is neither projective nor a summand of $\mathbb{Z}/4\mathbb{Z}$. $\mathbb{Z}$ is not semisimple since $2\mathbb{Z} \subset \mathbb{Z}$ is not a summand.

4 Forcing flat modules

Recall that a left module M is *flat* if $- \otimes_R M$ is exact. We use the standard criterion that M is flat if and only if $I \otimes_R M \to R \otimes_R M = M$ is injective for every finitely generated right ideal I.

Theorem 5. *Every left R-module is flat if and only if R is von Neumann regular.*

Proof. (⇒) Let $a \in R$. The module $M = R/Ra$ is flat. For a right ideal I, the map $I \otimes_R R/Ra \to R/Ra$ identifies, via $I \otimes_R R/Ra \cong I/Ia$, with the natural map $I/Ia \to R/Ra$ whose kernel is $(I \cap Ra)/Ia$. Injectivity for all I therefore means $I \cap Ra = Ia$. Take $I = aR$. Then $a \in aR \cap Ra = aRa$, so $a = axa$ for some x. Hence R is regular.

(⇐) Suppose R is regular. We first show that every finitely generated right ideal is generated by an idempotent. For one generator $a = axa$, the element $e = ax$ satisfies $e^2 = axax = ax = e$

and $aR = eR$ (as $a = ea$ and $e = ax \in aR$). For two generators a, b, pick an idempotent e with $eR = aR$ and an idempotent f with $fR = (1-e)bR$. Then $aR + bR = eR + fR$, because $b = eb + (1-e)b$ with $eb \in eR$. Writing $f = (1-e)by$ gives $ef = 0$. Put $g = e + f - fe$. Using $ef = 0$ one checks $eg = e$, $fg = f$ and $fe\,g = fe$, hence $g^2 = g$, and also $ge = e$, $gf = f$, so $e, f \in gR$. Since clearly $g \in eR + fR$, we get $eR + fR = gR$. Induction on the number of generators handles the general case.

Hence every finitely generated right ideal $I = eR$ is a direct summand of R_R. The inclusion $I \to R$ is then a split monomorphism, and tensoring with any left module M preserves split monomorphisms. So $I \otimes_R M \to M$ is injective for every M and every finitely generated I, and every M is flat by the criterion. □

Example 6. Every division ring and every product of fields (e.g. $\prod_n \mathbb{F}_2$) is regular. The ring $\mathbb{Q}$ is regular, and over it every module is flat, but $\mathbb{Z}$ is not regular: $\mathbb{Z}/2\mathbb{Z}$ is not flat over $\mathbb{Z}$ since it has torsion.

5 Hierarchy and further results

Combining the results above gives strict implications among the classes of rings:

$$\{\text{division rings}\} \subsetneq \{\text{semisimple rings}\} \subsetneq \{\text{von Neumann regular rings}\}.$$

The first inclusion is strict since $\mathbb{Q} \times \mathbb{Q}$ is semisimple but not a division ring; the second since $\prod_n \mathbb{F}_2$ is regular but not semisimple (it is not Noetherian). A semisimple ring is regular because each cyclic submodule is a summand and so is generated by an idempotent.

Table 1 summarizes all four forcing results together with further classical ones that we state without proof.

Every left R-module is…	**if and only if R is…**
free	a division ring (Thm. 2)
projective, or injective, or semisimple	semisimple (Thm. 3)
flat	von Neumann regular (Thm. 5)
Related results quoted from the literature	
every projective module is injective (equiv. injective ⇔ projective)	quasi-Frobenius [2]
every flat left module is projective	left perfect [1]
every submodule of a projective is projective	left hereditary [3]
every direct sum of injectives is injective	left Noetherian [1]

Table 1: Rings forcing a module type. The first three rows are proved above; the last four are classical theorems.

Remark 7. The last four rows are not forcing conditions in the strict sense used earlier, since they constrain a class of modules rather than force all modules to lie in it, but they have the same flavor: a statement about modules of a specific type is equivalent to an internal property of R.

References

[1] H. Bass, *Finitistic dimension and a homological generalization of semi-primary rings*, Trans. Amer. Math. Soc. **95** (1960), 466–488.

[2] C. Faith and E. A. Walker, *Direct-sum representations of injective modules*, J. Algebra **5** (1967), 203–221.

[3] H. Cartan and S. Eilenberg, *Homological Algebra*, Princeton University Press, 1956.

[4] T. Y. Lam, *Lectures on Modules and Rings*, Graduate Texts in Mathematics **189**, Springer, 1999.

[5] K. R. Goodearl, *Von Neumann Regular Rings*, 2nd ed., Krieger, 1991.